

Alpha Forge

Project Report

End Term Evaluation 2026

Systematic Alpha Generation via
On-Chain, Macro & Factor-Based ML Modelling

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1 Introduction

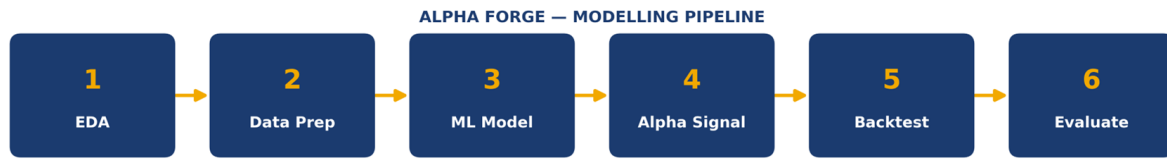
Project Premise

Alpha generation refers to earning returns in excess of a benchmark after adjusting for risk. Alpha Forge set out to test whether on-chain, macroeconomic, and fundamental/technical factor data can provide predictive insights into asset returns beyond traditional price-based analysis, and whether those insights survive contact with a disciplined, look-ahead-free backtest. Two markets were used as the testing ground for this question: **Bitcoin**, where the available signal is on-chain network activity and macro liquidity conditions, and the **Nasdaq-100**, where the available signal is company fundamentals, price momentum, and technical positioning. A single risk-and-backtesting toolkit — built first and shared across both — was applied consistently to both.

Project Objective: The objective of this project was to transform raw on-chain, macroeconomic, and fundamental/technical data into actionable trading signals across two distinct markets — Bitcoin and the Nasdaq-100 — by combining quantitative analysis, market risk estimation, and machine learning. The project began by building a common analytical foundation: the financial performance metrics, risk analytics (beta, CAPM, diversification), and machine learning fundamentals (loss functions, empirical risk, feature engineering) that any alpha-generation strategy needs to be evaluated honestly. That foundation was then applied twice over, first to construct and backtest an on-chain/macro Bitcoin strategy, and then to construct and backtest a fundamental-momentum-technical equity factor strategy on the Nasdaq-100 — each carried through to a full out-of-sample evaluation, including an explicit check of whether the result represents genuine alpha or simply a risk-factor tilt. This report presents the complete project framework, the shared analytical machinery that underpins both models, and the final data pipeline and results for each market in turn.

2 Project Flow

- Step 1. EDA** — Verify and explore all input features (distributions, stationarity, correlations, regime behaviour) for each asset universe.
- Step 2. Data Preparation** — Handle missing values, normalise features, build a chronological train/validation/test split with no look-ahead bias.
- Step 3. Modelling** — Fit ML models: gradient-boosted regression for Bitcoin log-return prediction; a multi-class classifier for Nasdaq-100 return-tercile prediction.
- Step 4. Alpha Signal** — Convert model outputs into a daily Long/Short signal (Bitcoin) or a rebalanced Long/Short factor book (Nasdaq-100).
- Step 5. Backtesting** — Simulate each signal on historical, out-of-sample data and evaluate statistically and financially, net of costs.
- Step 6. Feature Importance Analysis** — Interpret feature/factor importance and document which signals carried the most predictive power.



3 Financial Concepts & Performance Metrics

3.1 Trading & Performance Metrics

- ▶ **Long position:** Holding an asset expecting price to rise.
- ▶ **Short position:** Betting price falls (the model's signal decides which one is taken each period).
- ▶ **Alpha:** Excess return of the strategy over the relevant buy-and-hold benchmark, after accounting for transaction costs.
- ▶ **Sharpe:** Return earned per unit of volatility taken; the headline risk-adjusted performance metric for this task.
- ▶ **Sortino:** Differentiates harmful volatility from total overall volatility by using the asset's downside deviation instead of total standard deviation.
- ▶ **Maximum Drawdown:** The largest peak-to-trough decline in portfolio value — a capital-preservation lens, separate from average return.
- ▶ **Log Returns vs Simple Returns:** Log returns ($\ln(P_t/P_{t-1})$) are time-additive and closer to symmetric, which is why they are used as the prediction target instead of percentage change.
- ▶ **Transaction Costs:** Every position change erodes returns; a strategy that looks profitable gross of costs can lose its edge net of costs.
- ▶ **Benchmark:** The chosen market index (or buy-and-hold) serves as the reference portfolio. Consistently outperforming the benchmark after accounting for risk and transaction costs is the ultimate test of a strategy's success.



3.2 Key Metrics

Sharpe Ratio. The Sharpe Ratio measures risk-adjusted return:

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p}$$

where R_p is portfolio return, R_f is the risk-free rate, and σ_p is portfolio volatility. **Sortino Ratio.** The Sortino Ratio measures risk-adjusted return using downside deviation:

$$\text{Sortino} = \frac{R_p - R_f}{\sigma_d}$$

where σ_d is the downside volatility.

4 Market Risk and Portfolio Analytics

This section introduces the tools used to evaluate market risk and asset performance. Concepts such as Beta, CAPM, Alpha, and portfolio analytics help determine whether returns are justified by the level of risk undertaken.

4.1 Beta and the Security Market Line

Core Principle

“Risk cannot be eliminated, only measured and managed.” Beta serves as a primary measure of systematic market risk.

Beta measures the sensitivity of an asset’s returns to movements in the overall market:

$$\beta = \frac{\text{Cov}(R_{\text{stock}}, R_{\text{market}})}{\text{Var}(R_{\text{market}})}$$

Beta Value	Interpretation
$\beta = 1$	Moves exactly with the market
$\beta > 1$	Amplified moves (aggressive)
$\beta < 1$	Muted moves (defensive)

Plotting expected return against beta gives the Security Market Line (SML). Assets positioned above the SML are considered undervalued, as they offer higher returns than predicted by their level of risk, while assets below the SML may be overvalued.

4.2 CAPM and Alpha

CAPM Expected Return:

$$\text{Expected Return} = R_f + \beta \times (R_m - R_f)$$

where R_f is the risk-free rate and R_m is the market return. The simplest alpha is excess return: $\alpha_{\text{simple}} = R_{\text{actual}} - R_{\text{benchmark}}$. However, higher returns do not necessarily imply superior performance, as assets with greater market exposure may naturally earn higher returns due to increased systematic risk. CAPM-adjusted alpha measures performance after accounting for the return justified by an asset’s level of systematic risk:

$$\alpha_{\text{CAPM}} = R_{\text{actual}} - [R_f + \beta(R_m - R_f)]$$

A positive CAPM alpha indicates that an asset or strategy has generated returns beyond those predicted by its systematic risk exposure. As a result, alpha serves as a key measure of value creation and investment skill.

Figure A: Risk-Return Analytical Chain

Beta Calculation → CAPM Expected Return → Alpha Generation

Beta, CAPM, and alpha form a single chain — each step feeds the next. In the Nasdaq-100 section, this same chain is extended with the Fama-French 3-Factor model, since a single market beta is not sufficient to explain that portfolio’s returns.

4.3 Portfolio Construction and Diversification

A diversified portfolio combines assets from different sectors to reduce concentration risk. Portfolio beta is the weighted average of individual asset betas, while portfolio return is the weighted average of constituent returns.

Figure B: Diversification Mechanics

High Correlation (> 0.80)

Five stocks move together $\Rightarrow$ risks add up, high volatility

Low Correlation (< 0.20)

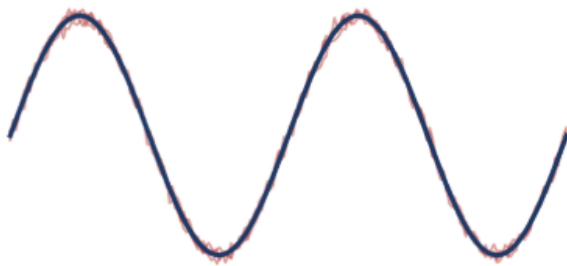
Swings cancel out $\Rightarrow$ smoothed portfolio curve

The Diversification Lesson. Diversification is effective only when portfolio assets are exposed to different sources of risk. Holding multiple highly correlated assets provides limited risk reduction, whereas combining assets from different sectors can significantly smooth portfolio volatility.

Illustrative example: A portfolio generating an 18% return with a $\beta = 1.6$ may have a CAPM-expected return of 16%, implying an alpha of only 2%. In contrast, the same 18% return achieved with a beta of $\beta = 0.7$ would indicate substantially stronger risk-adjusted performance.

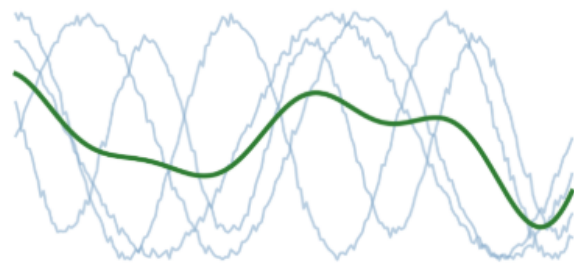
Same Number of Stocks, Different Diversification Benefit

All-Banking Portfolio
(stocks highly correlated)



Same swings repeated four times
 $\rightarrow$ risk barely reduced

Mixed-Sector Portfolio
(stocks weakly correlated)



Ups and downs cancel out
 $\rightarrow$ smoother, lower-risk line

5 Market Data Analysis & Rhode Features

5.1 The Nine Rhode Features

These nine engineered features formed the input space for the Bitcoin Alpha Generation model (Section 8).

Rhode ture	Fea-	Signal Type	What It Proxies
True_Market_Cap		Leading (valuation)	Exact network valuation; precise supply basis for all ratio features
FeeTotUSD		Leading (demand)	Real-time blockchain demand and congestion
Fee_to_MCap		Leading (valuation ratio)	Earnings-yield-style proxy: usage relative to valuation
DXY_Close		Lagging (macro)	Inverse gauge of global dollar liquidity
TNX_Yield		Lagging (macro)	Risk-free rate / discount-rate proxy
AdrActCnt		Leading (adoption)	Organic user growth vs. pure price speculation
HashRate		Lagging (security)	Miner conviction; implicit price-support level
Rolling Volatility		Lagging (risk)	Regime flag; normalises feature/position scaling
Log Returns		Target variable	$\ln(P_t/P_{t-1})$ — the variable all models predict

1. **True_Market_Cap** — price $\times$ exact circulating supply; more accurate valuation base than aggregator figures using lagged supply.
2. **FeeTotUSD** — total dollar fees paid to miners; the closest thing BTC has to “revenue,” reflecting real transactional demand.
3. **Fee_to_MCap** — fee $\div$ market cap; an earnings-yield-style ratio — rising values mean more usage per dollar of valuation.
4. **DXY_Close** — dollar index; inverse proxy for global liquidity, since a stronger dollar tightens credit and pressures BTC.
5. **TNX_Yield** — 10-year Treasury yield; risk-free rate / discount-rate proxy, raises the opportunity cost of holding BTC when it rises.
6. **AdrActCnt** — unique active addresses; Metcalfe’s-law-style adoption proxy that separates organic growth from pure speculation.
7. **HashRate** — total computational power securing the chain; sustained growth signals miner confidence (revealed preference via capex).
8. **Rolling Volatility** — std. dev. of recent log returns; flags regime shifts and normalises feature/position scaling.
9. **Log Returns** (target) — $\ln(P_t/P_{t-1})$, the variable all models predict.

5.2 Data Cleaning

- Missing macro values (DXY, TNX have weekend/holiday gaps that on-chain data doesn’t) handled via forward-fill only — never interpolation, since that would use future-known values.
- Normalisation: features span wildly different scales (trillions vs. 100 vs. single-digit % vs. exahashes); standardised for linear/regularised models, not required for tree-based models.
- Chronological train/validation/test split (e.g. 70/15/15 by date) — never random shuffle, since that would leak future information into training.

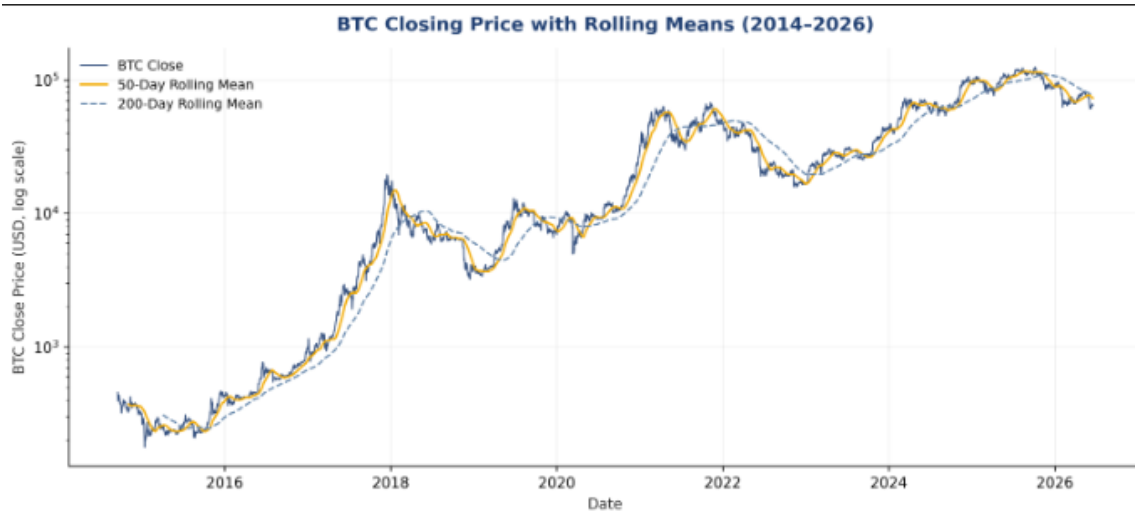
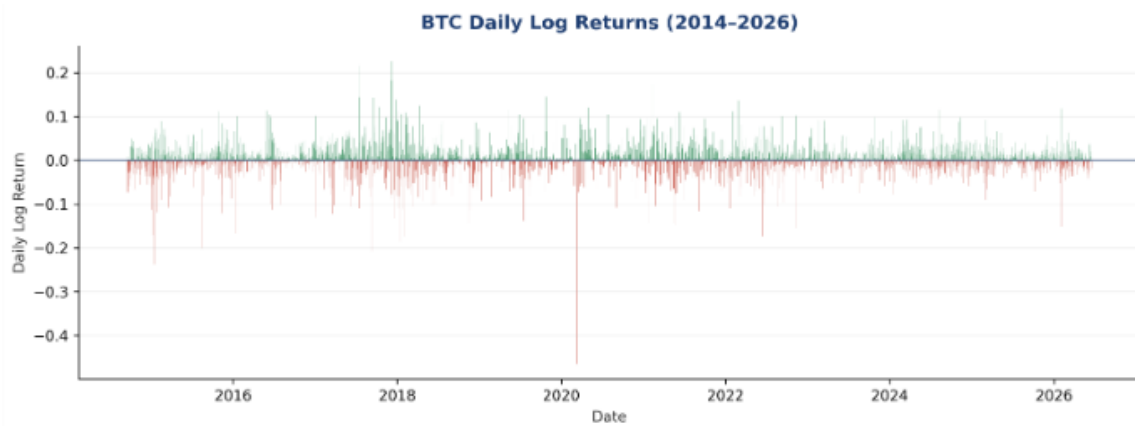
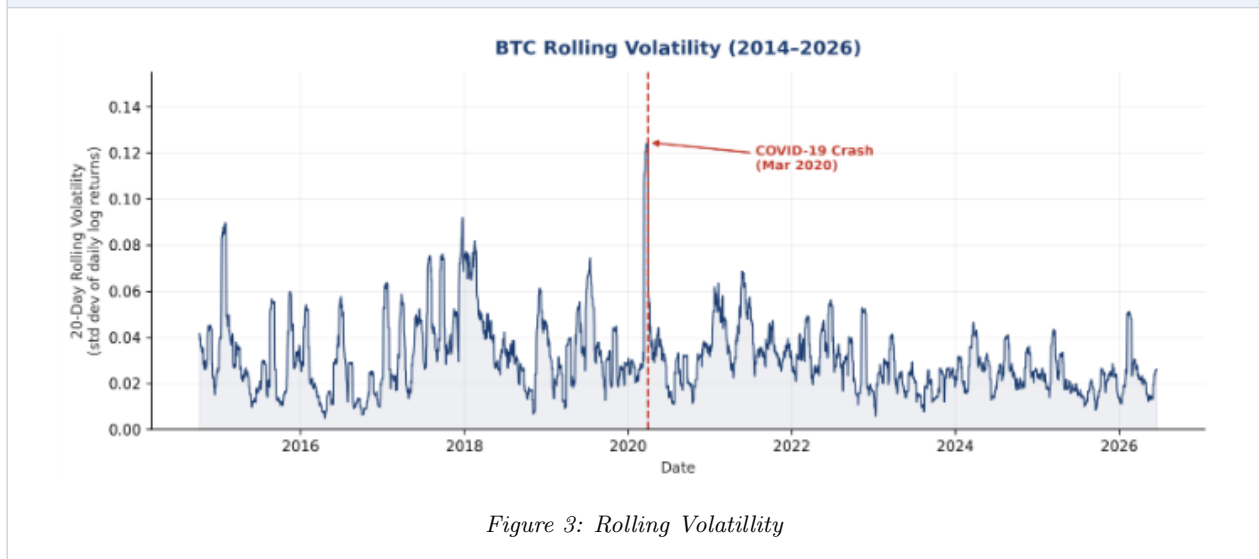
Figure: Bitcoin Price Chart*Figure 1: Bitcoin Price Chart***Figure: Log Returns Time Series***Figure 2: Log Returns Time Series*

Figure: Rolling Volatility

6 Machine Learning & Feature Engineering

This section steps back from trading-specific tools to the general ML ideas — loss, risk, optimisation, features — that both final deliverables (Sections 8 and 9) apply directly. Getting this math right is what makes those models trustworthy rather than a black box.

6.1 Loss Functions and Empirical Risk

Loss $\ell(y, \hat{y})$ measures how wrong one prediction is; it is always ≥ 0 .

Loss Function	Formula	Use Case
Zero-One	$\mathbf{1}[\hat{y} \neq y]$	Classification accuracy
Mean Squared Error	$\frac{1}{N} \sum (y_i - \hat{y}_i)^2$	Continuous returns
Cross-Entropy	$-\sum y_i \log \hat{y}_i$	Probability outputs

Empirical Risk Minimisation (ERM): Since no model sees the true distribution of all possible data, we calculate the average loss over the N examples we have and minimise it:

$$\hat{R}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(y_i, h_{\theta}(x_i))$$

Machine learning models are trained by minimising a loss function, which quantifies prediction error. The Bitcoin model (Section 8) minimises an MSE-family loss for continuous log-return prediction; the Nasdaq-100 model (Section 9) minimises a cross-entropy loss for tercile classification.

6.2 Linear Regression: The Closed-Form Solution

The linear model $h(x) = wx + b$ is the simplest and the basis for most others. Stacking parameters into θ and inputs into matrix X , the risk $J(\theta) = \frac{1}{N} \|y - X\theta\|^2$ forms a bowl-shaped (paraboloid) surface. Setting its gradient to zero gives a direct, one-step solution:

Normal Equation (Closed-Form Solution)

$$\theta^* = (X^{\top} X)^{-1} X^{\top} y$$

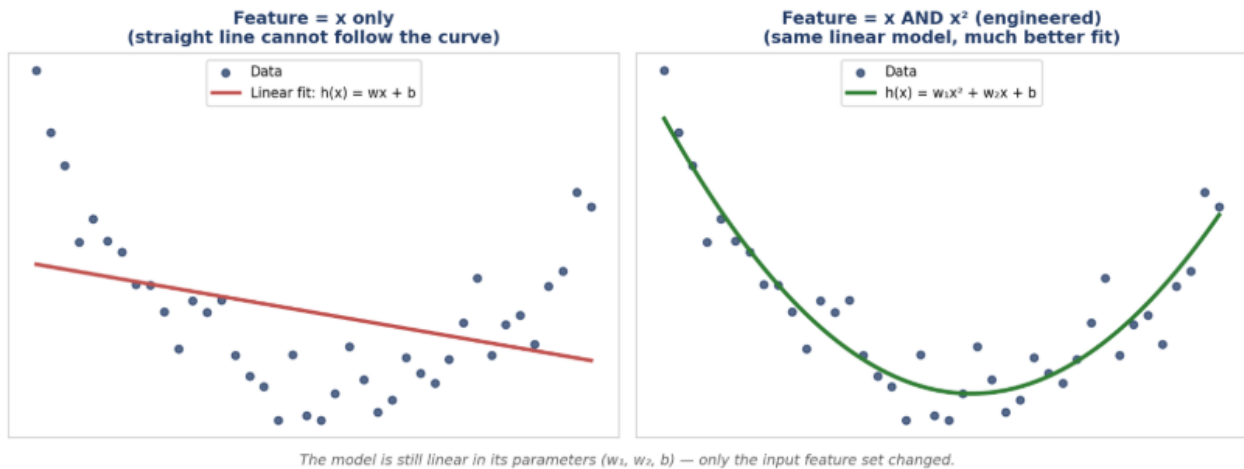
Requires $(X^{\top} X)$ to be invertible — the input features must not be redundant copies of each other.

6.3 Feature Engineering

Real relationships are rarely straight lines. Feature engineering transforms raw market, on-chain, and fundamental data into more informative predictors. Adding x^2 as an engineered feature lets the same linear model trace a curve.

Caution: Excessive feature engineering can lead to overfitting, where a model performs well on historical data but fails to generalise to unseen market conditions.

Feature Engineering: Helping a Linear Model See a Curve



Three model families carry this machinery into the two markets studied in this report:

- ▶ **LightGBM (gradient-boosted trees)** — used for Bitcoin log-return regression; captures non-linear interactions between on-chain and macro features without manual transforms.
- ▶ **XGBoost multi-class classifier** — used for Nasdaq-100 tercile classification; ranks stocks by predicted probability of landing in the top or bottom return tercile.
- ▶ **Linear / Fama-French regression** — used in the alpha-attribution step, to check whether ML-driven returns represent genuine skill or simply a factor tilt.

With the shared toolkit in place, the next two sections apply it end-to-end: first to Bitcoin, using the on-chain and macro Rhode Features introduced in Section 5, and then to the Nasdaq-100, using a fundamental-momentum-technical factor stack.

7 Quant Dashboard & Backtesting Framework

Objective

The dashboard deliverable turns the project's indicator and machine-learning strategies into a common, user-facing workflow: a ticker is supplied, market data are fetched through **yfinance**, a technical or ML strategy produces a signal, the shared backtester simulates execution, and the dashboard returns the equity curve, trade statistics, benchmark comparison, and live fundamentals.

7.1 End-to-End Architecture

The implementation is deliberately strategy-agnostic. Any component that produces a signal compatible with the backtester can use the same execution and evaluation engine.



7.2 Technical Strategies and Weighted Signals

Seven requested indicators — MACD, RSI, Bollinger Bands, ATR, Keltner Channel, OBV, and ADX — are implemented, with an SMA crossover as an additional indicator. Each indicator returns a

score on the $[-1, 1]$ scale. The weighted composite normalises the indicator weights to sum to one; the dashboard then converts the composite to a trading decision using a fixed threshold: above 0.3 is long, below -0.3 is short, and values in between are flat.

7.3 Backtesting and Look-Ahead Control

The backtester is bar-by-bar and strategy-agnostic rather than a simplified vectorised return calculation. Most importantly, the raw signal is shifted forward by one bar before trade sizing, so a signal formed from today's closing information can only affect the next available trading bar. Transaction costs and optional slippage are supported, along with multiple position-sizing modes and risk controls such as stop-loss, take-profit, trailing stops, and a maximum-drawdown kill switch. The engine computes a broad set of metrics including CAGR, Sharpe, Sortino, Calmar ratio, drawdown, win rate, profit factor, and trade activity.

7.4 Machine-Learning Integration

The ML layer is designed to reuse the same backtester. The submitted repository registers Logistic Regression, XGBoost, LightGBM, and an equal-weight ensemble, with engineered OHLCV-based features and a shared scaling pipeline. ML training uses a strict date-based split corresponding to an intended 80/20 train-test structure, while the backtester applies the same one-bar execution shift used for technical strategies. The project review also identifies two scope notes: a second regression-based model is still required to fully satisfy the original brief, and the current ML outputs are continuous, volatility-scaled scores in $[-1, 1]$ rather than discrete $+1/0/-1$ signals.

8 Bitcoin Alpha Generation: On-Chain & Macro ML Modelling

Objective

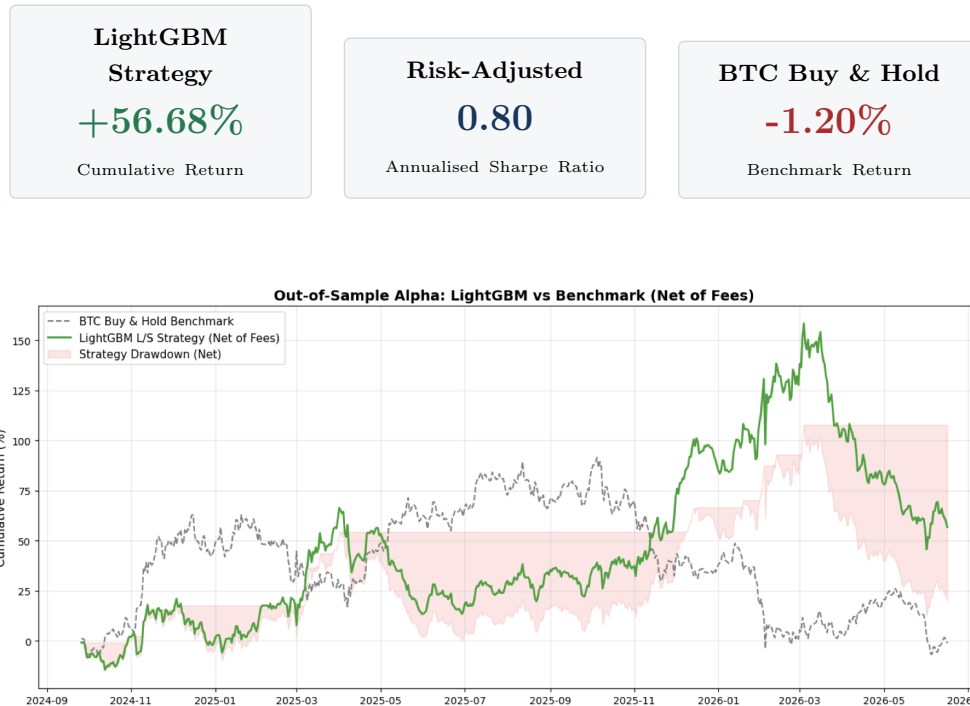
Can on-chain and macroeconomic information be used to generate predictive alpha in Bitcoin? Using 4,292 daily observations spanning September 2014–2026, the nine pre-engineered Rhode Features (Section 5) covering network activity, valuation, adoption, mining, volatility, liquidity, and interest rates were fed into a gradient-boosted model to predict daily log returns, chronologically split with no look-ahead bias, and converted into a daily Long/Short trading signal.

8.1 Learning Outcomes

- ▶ Developed an understanding of how on-chain and macroeconomic variables can influence Bitcoin returns.
- ▶ Learned to work with financial time-series data while avoiding look-ahead bias and ensuring chronological validation.
- ▶ Applied machine learning to return prediction and translated model outputs into actionable trading signals.
- ▶ Gained practical experience in strategy backtesting and risk-adjusted performance evaluation, using metrics such as Sharpe Ratio, cumulative return, and drawdown.
- ▶ Learned to interpret feature importance and connect model outputs with underlying financial and economic factors.

8.2 Out-of-Sample Results

The LightGBM-based strategy generated a **56.68%** cumulative gross return during the out-of-sample testing period, compared with -1.20% for Bitcoin Buy & Hold over the same window. The strategy achieved an annualised gross Sharpe Ratio of **0.80**, demonstrating positive risk-adjusted performance under the project’s look-ahead and transaction-cost controls. The equity curve below is the corresponding out-of-sample plot shown *net of fees*.



Overall, these results indicate that combining on-chain and macro signals with machine learning can produce meaningful out-of-sample alpha relative to passive Bitcoin exposure. The same question — does a systematic, ML-driven signal beat a passive benchmark once risk and discipline are accounted for? — is now asked of a very different asset class and information set in the Nasdaq-100 section.

9 Equity Factor Alpha: A Long-Short ML Portfolio on the Nasdaq-100

Objective

The same alpha-generation philosophy applied to Bitcoin was tested a second time in equities: a long-short factor portfolio on Nasdaq-100 stocks, ranked using fundamental, momentum, and technical factors together with a machine-learning classifier, backtested against an equal-weight baseline and a plain buy-and-hold benchmark, with alpha measured against the Fama-French 3-factor model.

9.1 Data

The universe is 102 Nasdaq-100 tickers, with daily OHLCV prices pulled via **yfinance** for January 2017 to December 2022. Of these, 94 tickers have a full price history across the whole window; the remaining late-IPO names are documented rather than silently dropped, since they simply don’t have enough history to compute yearly factors or long-window momentum. Annual fundamental

ratios (P/E, P/B, ROE, ROA, Debt/Equity, growth figures, and so on) were reshaped into a tidy year-by-stock panel, with the same annual value repeated across all twelve months of that year — so a stock’s 2021 P/E, for example, applies to every month in 2021 until the next year’s figure replaces it. Fama-French 3-factor monthly data (Mkt-RF, SMB, HML, RF) was taken from Kenneth French’s data library, starting January 2022 to line up with the backtest window.

9.2 Factors Built

Nine factors were built across six groups: valuation, profitability, growth, leverage/quality, momentum, and technicals. The technicals group was added to provide predictive information beyond price returns and accounting ratios.

Factor Group	Inputs Used	Direction
Valuation	P/E, P/B, EV/EBITDA	Cheaper is better
Profitability	ROE (proxy), ROA (proxy), Gross Margin	Higher is better
Growth	YoY EPS growth, YoY EBITDA growth	Higher is better
Leverage / Quality	Debt/Equity, Current Ratio, Asset Turnover	Lower D/E, higher current ratio and turnover is better
Momentum (mandatory)	1, 3, 6, 12-month trailing returns	Higher is better
Technicals	EMA/HMA distance, MACD, Bollinger %B	Signal-dependent

Every factor is sign-adjusted first, so a higher value always means “more attractive” regardless of the raw direction, and is then converted into a percentile rank across all stocks on each month-end date. This puts every factor on the same 0–1 scale before it is fed into the model, and it is what each rebalance date’s ranking is based on.

9.3 ML-Based Ranking

Target variable: next month’s return bucket — bottom, middle, or top tercile — rather than a raw return number. This is a classification approach: instead of predicting an exact return, the model predicts how likely a stock is to land in each tercile next month. **Features:** 21 candidate factor ranks in total, covering fundamentals, momentum, and technicals. SHAP importance on the training set was used to narrow this down to the 7 features that mattered most:

- ▶ 12-month momentum rank
- ▶ HMA (Hull moving average) distance rank
- ▶ P/E rank
- ▶ 6-month momentum rank
- ▶ P/B rank
- ▶ EMA-26 distance rank
- ▶ 1-month momentum rank

Model: an XGBoost multi-class classifier (300 trees, max depth 3, learning rate 0.05), trained on January 2017–December 2021 and tested only on 2022 — a strict chronological split with no look-ahead. As with a return-regression approach, momentum and price-based signals dominate

the top of the importance ranking (12-month momentum and HMA distance are the two strongest single features), with fundamentals such as P/E and P/B contributing meaningfully but trailing behind the price-based features. Leverage and asset-turnover ranks sit at the bottom of the list, contributing the least.

9.4 Method for the Final Ranking and Portfolio

At every rebalance date, the top 3 stocks by predicted probability of landing in the top tercile were held long, and the bottom 3 by predicted probability of the bottom tercile were held short. Position weights within each leg were set dynamically rather than equally: model confidence scores were exponentially reweighted, so a stock the model was more confident about received a larger position within its leg. Positions were held for 2 months, then closed and re-picked at the next rebalance — 6 non-overlapping rebalances across 2022 in total. The signal itself is built at month-end close and evaluated over the following 2-month window, so there is no look-ahead in either the ranking or the return that gets measured against it. Two comparisons were used as benchmarks alongside the dynamic-weight portfolio: an equal-weight version of the same top-3/bottom-3 picks, and an equal-weight buy-and-hold portfolio across all Nasdaq-100 names, plus the Fama-French 3-factor model for the FF3 alpha-attribution subsection.

2022 Realised Stock Ranking (Sanity Check)

Applying the same top-3/bottom-3 logic to the year’s actual outcomes:

Top 3 — Long	Realised Return	Bottom 3 — Short	Realised Return
PDD	+45.4%	ROST	-14.4%
VRTX	+29.8%	MAR	-15.9%
GILD	+23.7%	SBUX	-17.5%

The same top-3/bottom-3 logic is applied three times in the pipeline: to the fundamental composite score at each year-end, to the model’s predicted tercile probabilities at each 2-month rebalance, and — as shown above — to realised returns after the fact, to sanity-check how the picks actually performed.

9.5 Evaluation: ML-Ranked vs Equal-Weighted vs Buy-and-Hold

Metric	Dynamic-weight ML	Equal-weight ML	Nasdaq-100 Buy & Hold
Total return (2022)	+25.3%	+28.5%	-24.5%
Sharpe ratio	1.92	2.85	-0.81
Max drawdown	-2.9%	-0.4%	-31.1%

2022 was a down year for the Nasdaq-100, and both long-short ML portfolios comfortably beat a plain buy-and-hold, which lost close to a quarter of its value with a drawdown over 31%. Between the two ML versions, equal-weight actually came out ahead of the dynamic-weight version on both total return and Sharpe ratio. This is mostly because the confidence-based weighting concentrated capital into a smaller number of names during a volatile stretch of the year, which added risk without a matching pickup in return.

9.6 Alpha vs Fama-French 3-Factor Model

Regression:

$$R_{\text{portfolio}} - R_f = \alpha + \beta_{\text{MKT-RF}}(\text{MKT} - \text{RF}) + \beta_{\text{SMB}}(\text{SMB}) + \beta_{\text{HML}}(\text{HML}) + \varepsilon$$

estimated on the six 2-month holding periods of 2022.

Portfolio	Annualised α	α p-value	β_{MKT}	β_{SMB}	β_{HML}	R^2
Dynamic-weight	-16.4%	0.69	0.37	7.34	2.77	0.72
Equal-weight	-3.0%	0.84	0.28	4.86	2.02	0.94

Read honestly, neither portfolio shows a statistically significant alpha against FF3 — both p-values are far above the usual 0.05 threshold, and there are only a handful of 2-month observations in the 2022 test window to estimate the regression from. Both portfolios also load heavily on SMB (small-minus-big), well above a beta of 1, which means most of the return is coming from a size and style tilt rather than genuine stock-picking skill. For context, 2022's FF3 factor backdrop was Mkt-RF -21.0% , SMB -7.4% , and HML $+31.7\%$ — a year where value strongly outperformed growth, which lines up with why factor tilts explain so much of the portfolio's moves. A longer test period, spanning more than one year, would be needed to get a reliable read on whether either portfolio has real alpha.

9.7 Section Conclusion

The pipeline runs end to end: nine factors across four fundamental groups, a dedicated momentum group, and a technicals group, ranked as percentiles at every month-end; an XGBoost multi-class classifier that predicts each stock's next-month return tercile from the 7 most important of 21 candidate features; a confidence-weighted top-3/bottom-3 long-short book rebalanced every 2 months; and an FF3 alpha regression to check whether the results hold up against a standard risk model. Over the 2022 test period, both the dynamic-weight and equal-weight versions of the ML portfolio clearly outperformed a plain Nasdaq-100 buy-and-hold, which is the headline result. Between the two, equal-weighting turned out to be the better risk-adjusted choice this year. Momentum and price-based technical indicators (12-month momentum and HMA distance especially) were the strongest signals feeding the model, ahead of any single fundamental ratio. Neither portfolio shows a statistically significant alpha over the Fama-French 3-factor model, and both lean heavily on a size (SMB) tilt rather than pure stock-picking skill — a limitation of a single-year, six-rebalance test window rather than a failure of the pipeline itself.

10 Key Learnings

“A number is only half a story until it is interpreted.”

10.1 Reading Returns Honestly

The most important learning from Alpha Forge, reaffirmed by both the Bitcoin and Nasdaq-100 results, was that financial analysis needs both numbers and judgement. A high return looks attractive on its own, but it means something different at an 8% maximum drawdown than at 40%, and a 25.3% equity return means something different once it is regressed against a factor model and found statistically insignificant. The same headline figure can describe two completely different risk (or skill) profiles — which is exactly why this project paired every return number with a volatility, beta, drawdown, or factor-regression reading rather than reporting it alone.

10.2 Backtesting Discipline

A second learning sits underneath the first: a backtest is only as honest as the rule that built it. Executing a signal at day t 's close instead of day $t+1$'s open quietly hands the strategy information

it would never have had in real time, and the resulting curve stops describing anything that could actually be traded. Transaction costs carry a similar warning — a strategy that looks profitable gross of costs can lose much of that edge once every trade pays its toll, particularly for higher-frequency rules like the daily Bitcoin signal.

Why Risk-Adjusted Comparison Matters

A mean-reversion strategy and a trend-following strategy can each report the same 20% back-tested return and still not be comparable — one might earn it with an 8% drawdown and a Sharpe ratio near 1.4, the other with a 35% drawdown and a Sharpe ratio under 0.5. The return line alone cannot tell them apart. The Nasdaq-100 factor portfolio made the same point from a factor-attribution angle: two portfolios with similar headline returns can differ entirely in *where* that return came from.

Raw Return → Risk Check → Cost Check → Bias Check → Factor Check → Trustworthy Result

A backtested result only becomes a trustworthy result once it is checked against risk, cost, bias, and (where equities are involved) factor exposure — each step narrows what the return is actually allowed to claim.

10.3 Diversification and Factor Exposure Are About Movement, Not Headcount

The portfolio-construction concepts in Section 4 apply just as much at the factor level as at the stock level. Two individually risky stocks can lower portfolio risk if they are not highly correlated, while several individually “good” stocks can quietly create concentration risk if they all move together. At the factor level, a portfolio’s SMB and HML betas can dominate its return even when every individual stock pick looked sound — diversification and attribution are both decided by *exposure*, not by counting tickers or celebrating a headline number.

10.4 Skill vs. Style: A Central Lesson of This Project

Perhaps the sharpest lesson to emerge once the equity factor model was in place is the gap between a strategy that *looks* skilled and one that is *statistically* skilled. The Bitcoin model’s Sharpe of 0.80 and 56.68% out-of-sample return stand on their own as a single-asset, single-benchmark result. The Nasdaq-100 portfolio’s stronger raw numbers (up to +28.5% in a down year) look even more impressive at first glance — until the FF3 regression shows an insignificant alpha p-value and a heavy SMB/HML tilt. Together, the two results teach that outperforming a benchmark and generating genuine, statistically defensible alpha are not the same claim, and that the second one requires more data and more rebalances than a single test year can supply.

Lesson	Why It Matters
Clean data is the base of every reliable result	Errors at the data stage propagate into every return, volatility, beta, and factor loading calculated downstream.
Returns must be read with volatility and drawdown	The same headline return can describe very different risk profiles.
Backtests must avoid look-ahead bias and price in costs	A rule that “wins” on unrealistic execution proves nothing about live trading.
Diversification depends on correlation, not stock count	Five correlated stocks can behave like one concentrated bet.
ML models are only as good as their loss function and features	The wrong loss function or noisy features optimise for the wrong target entirely.
Outperformance is not the same as statistically significant alpha	A strong equity curve can still fail a factor-model significance test, especially with few rebalance observations.

11 Limitations and Future Work

11.1 Why a Backtest Can’t Promise the Future

Every result in this report is built on historical data, so none of it can guarantee future performance. Market structure, liquidity, transaction costs, and volatility regimes all shift over time, and a rule fitted to one regime carries no promise that it will survive the next one. The backtested window and the unseen regime that follows it are not equivalent, even when both came from the same underlying market.

11.2 Where CAPM, FF3, and Indicators Fall Short

CAPM has a narrower limitation: it explains expected return using only market risk (β), while real stock returns are shaped by many other factors it does not capture — which is exactly why the Nasdaq-100 portfolio was checked against a 3-factor model instead of CAPM alone. But even FF3 is not the final word: with only six 2-month rebalance observations in 2022, the FF3 alpha-attribution subsection is statistically under-powered, and neither portfolio’s alpha estimate can be trusted with confidence. Technical indicators carry a related weakness — most are lagging by construction, built from past prices, so they confirm a trend only after it has already begun.

Lagging Indicators	Backtest Window / Sample-Size Risk
Signals confirm a trend after it has already begun — the move is partially over by the time the model acts.	The backtested regime and the future regime are not the same; a single test year with six rebalances says little about what comes next.

11.3 What’s Next

Several extensions would tighten the gap between a backtested result and a live-tradeable one:

- ▶ **Multi-year backtesting** for the Nasdaq-100 factor model, to obtain enough rebalance observations for a reliable alpha estimate.
- ▶ **Monte Carlo simulation** to stress-test both strategies across many possible paths, not just the one that happened.
- ▶ **Regime-wise correlation and volatility forecasting**, since both shift over time rather than staying fixed.
- ▶ **Walk-forward testing** and portfolio optimisation with realistic constraints for both the Bitcoin and Nasdaq-100 signals.
- ▶ **Extending the factor model** beyond FF3 (e.g. adding a quality or low-volatility factor) to better explain the residual return in the Nasdaq-100 portfolio.
- ▶ **Cross-asset combination**: exploring whether the on-chain/macro Bitcoin signal and the equity factor signal are diversifying with respect to each other in a combined book.

12 Conclusion

Alpha Forge connected finance theory, Python implementation, and practical interpretation across data, indicators, backtesting, risk, portfolio construction, and machine learning, applying one shared analytical foundation to two complete, out-of-sample-tested trading systems. The project was never only about producing charts or running formulas — it was about understanding *why* each metric matters and how it supports an actual trading or investment decision.

Two Threads Running Through Everything

- ▶ Every later step depends on an earlier one being done honestly — a mis-cleaned price series, a look-ahead-biased signal, or an under-powered factor regression corrupts every beta, alpha, and Sharpe ratio calculated downstream.
- ▶ A number is only useful once it has been interpreted in context, whether that context is a drawdown, a correlation, a loss function, or a factor-model p-value.

Read together, the key learnings, the limitations, and the results describe the same project from three angles: what was understood, what it cannot yet promise, and what was actually built. The Bitcoin model demonstrates that on-chain and macro signals, combined with disciplined ML and backtesting, can produce meaningful out-of-sample alpha against a single-asset benchmark. The Nasdaq-100 factor portfolio demonstrates that a similar pipeline can be built for equities and can comfortably outperform the buy-and-hold benchmark on raw and risk-adjusted terms in a difficult year — while also demonstrating, through the FF3 regression, exactly how much scrutiny is required before “beat the benchmark” can be upgraded to “genuine alpha.” Overall, the project gives a solid, honestly-tested foundation for building more advanced quantitative trading systems in the future.

13 Deliverables Completed

Component	What Was Built
Indicators and Back-testing	Five indicators (SMA, EMA, RSI, MACD, Bollinger Bands) built from scratch, powering an EMA-crossover backtest with trade execution and performance metrics.
Quant Dashboard & Backtesting	A Django dashboard integrating live <code>yfinance</code> data, seven requested indicators, weighted signals, ML strategies, a shared backtester, equity curves, fundamentals, and a broad performance-metrics panel.
Beta, CAPM and Alpha	Beta, the SML, both simple and CAPM-adjusted alpha, and a five-stock multi-sector portfolio with weighted beta and return.
ML Foundation	Loss functions, ERM with the closed-form derivation, and feature engineering — the shared groundwork behind both models below.
Bitcoin Alpha Generation	Nine Rhode on-chain/macro features, a LightGBM return-prediction model, a daily Long/Short signal, and an out-of-sample backtest achieving +56.68% cumulative return and a 0.80 Sharpe ratio vs. a -1.20% BTC buy-and-hold benchmark.
Factor-Based Portfolio	ML Nine fundamental/momentum/technical factors, a SHAP-selected 7-feature XGBoost tercile classifier, a confidence-weighted top-3/bottom-3 long-short Nasdaq-100 book, and an FF3 alpha regression — outperforming buy-and-hold by over 50 percentage points in 2022 while honestly flagging insignificant alpha.

Two Unifying Threads

- ▶ Every later step depends on an earlier one being done correctly — a mis-cleaned price series corrupts every beta calculated downstream.
- ▶ A number is only useful once interpreted — beta without alpha, accuracy without a loss function, or a strong equity curve without a factor-model check is always half a story.

Together, this closes the loop the project opened with raw OHLCV and on-chain data: a shared risk-and-backtesting toolkit was built once, then used to construct, test, and honestly critique two independent trading systems — each checked against the risk, and where relevant the factor exposure, that produced its returns.